

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 2nd Year 2nd Term Examination, 2018

CE 2241

(Principles of Environmental Engineering)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

1. (a) What are the objectives of water supply? Mention the elements of water supply system with neat sketches. (07)
(b) Define- (i) Potable water (ii) Palatable water (iii) Wholesome water (06)
(c) Enumerate the general considerations for planning and design of a water supply system. (12)
(d) Distinguish between aquifer and aquitard. A 150 mm diameter tube well produces 100 lps with a drawdown of 3 m and a circle of influence of 120 m in diameter. The static depth of water in the tube well is 40 m. Calculate the co-efficient of permeability of the aquifer in which the tube well is sunk. (10)
2. (a) What are the basic requirements of drinking water? Write down the factors affecting per capita water consumption. (10)
(b) Why population forecasting is essential in water supply project? What are the methods for predicting population for a design period? (05)
(c) Write the Bangladesh standards and WHO guideline values of the following water quality parameter- (07)
(i) Arsenic (ii) Iron (iii) Chloride (iv) Color (v) Hardness (vi) Turbidity (vii) TC and FC
(d) The population data of a village is supplied below- (13)

Year	1940	1950	1960	1970	1980	1990
Population (Thousands)	20	25	28	32	37	43

Calculate the population in 2010 and 2050 by least square parabola method?
3. (a) Show the efficiency of a settling tank is dependent on surface area, not depth of tank. (09)
(b) What are the factors influencing settling velocity of particles in water? Define- discrete settling, hindered settling and flocculent settling. (11)
(c) What is coagulation? Which are the widely used coagulants? (06)
(d) Calculate the dimension of a rectangular settling tank to treat 100 m³ of water per hour when the overflow rate is 0.75 m/hr and the detention time is 2 hr. (09)
4. (a) What is disinfection? Explain the process of disinfection by chlorination. (12)
(b) Write down the name of different methods of arsenic removal from drinking water. (05)
(c) Define- (i) Chlorine demand (ii) Double Chlorination (iii) Break point chlorination and (iv) Super chlorination. (08)
(d) What are the main purposes of the construction of water transmission and distribution pipe lines? (04)
(e) What are the problems in ground water development in Bangladesh? (06)

Section – B

5. (a) What is sanitation system? Distinguish between the meaning of the terms of sewage and sullage. (08)
- (b) Give reason why various sanitation systems are based on water consumption level and population density. (10)
- (c) What are the important factors for sanitation in Bangladesh? (06)
- (d) In an effort to discourage people from open defecation the local authority in a village offers pre-cast concrete rings of 1.0 m diameter and concrete slabs to cover the pits at a very stabilized rate. Design a simple pit latrine for an average family of 7 persons who uses water for anal cleansing. The ground water table is below 5 m and latrine is to serve the family for at least 4 years. (11)

6. (a) What is simple pit latrine? What are the general design considerations for pit latrines? (10)
- (b) Give reason why volume in wet pit latrine is less than in dry pit latrines. (05)
- (c) What is ROEC? What is difference between compost latrines and VIP latrines? (10)
- (d) Name the factors which affect the performance of septic tanks. Design a septic tank to serve a household of 10 persons who produce 90 lcpd of wastewater. The tank is to desludged every 3 years. (10)

7. (a) What is the difference between separate sewerage system and combined sewerage system? (05)
- (b) What is self-cleansing velocity and waste stabilization ponds? Describe the symbiosis between algae and bacteria. (10)
- (c) What are the factors bring in significant cost savings in small bore sewerage (SBS) systems? Comment on the applicability of SBS in Bangladesh. (10)
- (d) What are the effects of solid waste mismanagement? Briefly describe the various collection systems of solid wastes. Which type is the most appropriate for Bangladesh and why? (10)

8. (a) What is pollution and pollutant? Briefly describe different types of pollutants. (10)
- (b) What is water and soil pollution? What are the causes and control measures may be taken for soil pollution? (10)
- (c) What are the typical design elements for storm drainage? (05)
- (d) The catchment area is of 300 hectares. The surface cover in the catchment can be classified as given in Table. (10)

Type of cover	Co-efficient of Runoff	Percentage (%)
Roofs	0.90	15
Pavements and yards	0.80	15
Lawns and gardens	0.15	25
Roads	0.40	20
Open ground	0.10	15
Single family dwelling	0.50	10

Calculate the runoff co-efficient and quantity of storm water runoff, if intensity of rainfall is 30 mm/hr for rain with duration equal to the time of concentration. If population density in the area is 350 persons/hectre and rate of water supply is 200 lpcd, Calculate design storm discharge for separate system.

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
 BURP 2nd Year 2nd Term Examination, 2018
Hum 2217
 (Public and Local Government Finance)

Full Marks: 210

Time: 3.0 hrs

- N.B.: i) Answer any **three** questions from each section in separate script
 ii) Figures in the right margin indicate marks
 iii) Assume data if necessary

Section - A

Marks

1. (a) Define public finance. What are the points of difference between public and private finance? (15)
- (b) What is the free-rider problem? Why does the free-rider problem induce the government to provide public goods? How can the government solve the problem? (20)
2. (a) What is externality? Greater consumption of alcohol leads to more motor vehicle accidents and thus imposes costs on people who do not drink and drive. Illustrate the market for alcohol labeling the demand curve, the social-cost curve and the market equilibrium level of output. (20)
- (b) On your graph, shade the area corresponding to the deadweight loss of market equilibrium. Explain. (15)
3. (a) Draw the supply and demand curve for cookies. If the government imposes a tax on cookies, show what happens to the quantity sold, the price paid by buyers and the price paid by sellers. In your diagram, show the deadweight loss from the tax. Explain the meaning of deadweight loss. (20)
- (b) How do the elasticities of supply and demand effect the deadweight loss of a tax and surplus of consumers and producers? (15)
4. (a) Briefly discuss about local government structure in Bangladesh. (20)
- (b) Discuss about the composition and function of Union Parishad in Bangladesh. (15)

Section – B

5. (a) What are the objectives of public expenditure? Discuss. (15)
- (b) Explain the principles of public expenditure. (10)
- (c) Show the effect of public expenditure on national income. (10)
6. (a) Distinguish between development budget and revenue budget. (10)
- (b) Discuss the role of Government Budget. (10)
- (c) Explain methods of formulating budget in a country like Bangladesh. (15)
7. (a) What are the relation between development and plan? Explain. (15)
- (b) What is deficit financing? (05)
- (c) Discuss role of city corporation in Bangladesh for giving standard living facilities of its citizen. (15)
8. Write short notes on the following:
 - (a) Public debt and budget deficit (15)
 - (b) Common function of local government (10)
 - (c) Decentralization (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Regular Examination, 2018.

URP 2281
(GIS and Remote Sensing)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) GIS can be applied to solve many practical problems with respect to location, condition, trend, routing, pattern, and modeling aspects. Take a planning spatial problem to solve with those equations. (15)
- (b) Explain the data model with example in GIS. (10)
- (c) "The evolution of geotechnology community has broadened its membership in members, interests, background and depth of understanding" – explain with necessary example. (10)
2. (a) Why GIS as a separate profession is important and so hot? Explain with a focus to Bangladesh. (10)
- (b) Explain the equation, "Spatial Reference = Datum + Projection + Coordinate System" with necessary example. (15)
- (c) Explain the different types of error produced while digitizing with necessary figures. (10)
3. (a) Explain three rules of topology with necessary sketches. (10)
- (b) Explain the different types of spatial queries / operations and applicability in solving spatial problems using GIS. (15)
- (c) Write down the importance of layers in GIS with a focus to the practical applications in urban planning. (10)
4. (a) Write down the functions of ArcMap, ArcCatalog, ArcScene, ArcGlobe and ArcTool box. (10)
- (b) Explain the visual balance of map layout. Write down some tricks to use for making better maps. (10)
- (c) Describe the applications of Network Analyst in Urban planning especially solving urban spatial problems. (15)

Section – B

5. (a) Describe remote sensing satellite with essential example. (05)
- (b) What do we measure in Remote Sensing in case of urban and regional planning, environmental planning, and transportation planning? Briefly explain with relevant example. (10)
- (c) Write down your understanding about the need for remote sensing and photogrammetric education. (10)
- (d) Differentiate among Ground-based, Airplane-based, Satellite-based remote sensing platforms based on their applications with necessary example. (10)

6. (a) "A common radiometric response is required for quantitative analysis of multiple satellite images of a scene acquired on different dates with different sensor." – are you agreed with the statement? State your rationale with your answer. (15)

- (b) Compare between two sets of satellite Images having different types of resolutions in the following. Explain with proper example. (10)

Satellite Image (A)	Satellite Image (B)
• High spatial resolution (0.5 - 4 m) • Low spectral resolution (1- 3 bands) • High radiometric resolution (16 bit and above)	• High spatial resolution (0.5 - 4 m) • High spectral resolution (16- 220 bands) • Medium radiometric resolution (8 bit image)

- (c) When do you use True Color Combination (TCC) and False Color Combination (FCC) in remote sensing analysis and why? (10)

7. (a) Explain the relationship between photo scale and map scale. (05)

- (b) "Photography is the first part of photogrammetry where metrology is the second part" – Briefly explain with proper example. (10)

- (c) Explain the different types of aerial photographs with essential sketches and examples. (10)

- (d) Assume that the distance between two points was measured to be 130 mm on a vertical photograph and 350 mm on a map. If the surveying ground distance between the same two points is 7000 m, what will be the photo scale and map scale and the scale reciprocals of the photograph and the map? Provide your judgments on the photo scale and map scale to describe the level of detailing of the photographs. (10)

8. (a) Differentiate between Photogrammetry and Remote Sensing technology with necessary examples. (05)

- (b) Briefly explain the different types of scale of a vertical photograph by drawing essential figures. (10)

- (c) Explain any four applications of the Photogrammetry of the following: (20)

- (i) Topographic Mapping (ii) Cadastral Plans (iii) Land use maps (iv) Exploration and Reconnaissance (iv) Detailed Survey of Historic Buildings

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Regular Examination, 2018
URP 2291
(Statistics for Planners-II)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Necessary chart and graphs will be supplied, t-test, z-test and χ^2 -test.
iv) Assume necessary data.

Section – A

1. (a) What is meant by correlation? Discuss the types of correlation with appropriate examples. (08)

- (b) Find the coefficient of correlation between the Age and Income earned from the following table: (15)

Age Group	Income Earned (in BDT)				
	10,000	20,000	30,000	40,000	50,000
20-30	4	6	3	7	1
30-40	2	8	15	7	1
40-50	3	9	12	6	2
50-60	8	4	2	-	-

- (c) An examination of eight applicants for a clerical post was taken by a firm. From the marks obtained by the applicants in the Accountancy and Statistics papers, compute rank coefficient of correlation. (12)

Applicant	A	B	C	D	E	F	G	H
Marks in Accountancy	15	20	28	12	40	60	20	80
Marks in statistics	40	30	50	30	20	10	30	60

2. (a) What is regression? Mention the properties of regression coefficients. (10)

- (b) Prove that "Regression coefficients are independent of change of origin but not scale". (10)

- (c) Define coefficient of determination. In trying to evaluate the effectiveness in its advertising campaign, a firm compiled the following information: (15)

Year	1996	1997	1998	1999	2000	2001	2002	2003
Adv. Exp. (In thousand BDT)	12	15	15	23	24	38	42	48
Sales (Lakh BDT)	5.0	5.6	5.8	7.0	7.2	8.8	9.2	9.5

Calculate the regression equation of sales on advertisement expenditure.

3. (a) What is time series analysis? Illustrate the components of time series with necessary examples. (10)

- (b) Explain semi-average method for measuring trend. Mention the merits and limitations of semi-average method. (10)

- (c) The sales of a company in lakhs of BDT for the year 1999 to 2005 are given below: (15)

Years	1999	2000	2001	2002	2003	2004	2005
Sales (m. tones)	32	47	65	92	132	190	275

Estimate sales figure for the year 2008 using an equation of the form $Y = ab^X$, where X = years and Y = sales.

4. (a) What are the limitations of coefficient of correlation? (05)

- (b) What are regression lines? Exemplify how they help in decision making. (08)

- (c) The following results of the height (X) and weight (Y) of 1,000 managers are given, (10)

Mean (X) = 68.00", Mean (Y) = 158 lbs, Standard Deviation (X) = 2.50"
Standard Deviation (Y) = 20 lbs

Coefficient of correlation between X and Y = 0.6. Estimate from the above data the height of a manager whose weight is 200 lbs.

- (d) Determine the equation of the straight line which best fits the following data: (12)

X	10	12	13	16	17	20	25
Y	19	22	24	27	29	33	37

Section – B

5. (a) What is hypothesis? Discuss the procedures involved in testing of hypothesis. (12)
 (b) Distinguish between One-tailed and Two-tailed tests with necessary examples. (10)

- (c) You are working as a purchase manager for a company. The following information has been supplied to you by two manufacturers of electric bulbs: (13)

	Company A	Company B
Mean life (in hours)	1300	1288
Standard deviation (in hours)	82	93
Sample size	100	100

Which brand of bulbs are you going to purchase if you desire to take a risk of 5%?

6. (a) What do you mean by test of hypothesis? (05)
 (b) Draw your concept on any three: (15)
 (i) Null hypothesis (ii) Degrees of freedom (iii) Level of significance
 (iv) Critical region.
 (c) The manufacturer of a certain type of electric bulbs claims that his bulbs have a mean life of 25 months with a standard deviation 5 months. A random sample of such bulbs gives the following value: (15)
 Life in month: 24, 26, 30, 20, 20, 18. Can the producer claim to be valid at 1% level of significance?

7. (a) What is meant by degree of freedom? State the properties of a t-distribution. (10)
 (b) Prices of shares (in BDT) of a company on the different days in a month were found to be: (13)
 66, 65, 69, 70, 69, 71, 70, 63, 64 and 68
 Test whether the mean price of the shares in the month is 65.

- (c) Ten persons were appointed in an officer cadre in an office. Their performance was noted by giving a test and the marks were recorded out of 100. They were given 4 months training and a test was held and marks were recorded out of 100. (12)

Employee	A	B	C	D	E	F	G	H	I	J
Before Training	80	76	92	60	70	56	74	56	70	56
After training	84	70	96	80	70	52	84	72	72	50

By applying the t-test, can it be concluded that the employees have benefited by the training?

8. (a) What is Chi-square Test? Mention the important properties of Chi-square Distribution. (10)
 (b) A certain drug is claimed to be effective in curing cold. In an experiment on 500 persons with cold, half of them were given the drug and half of them were given the sugar pills. The patients' reactions to the treatment are recorded in the following table: (15)

	Helped	Harmed	No effect	Total
Drug	150	30	70	250
Sugar Pills	130	40	80	250
Total	280	70	150	500

On the basis of this data, can it be concluded that there is a significant difference in the effect of the drug and sugar pills?

- (c) Distinguish between sample and population. Mention the applications of Chi-square Test in urban planning. (10)
